Benjamin Zhao

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TECHNICAL SKILLS

Languages: C++, Python, Java, SQL, Go
Systems & Infrastructure: Linux, Computer Architecture, Virtualization, OCI, Docker, Terraform
Development & Reliability: CI/CD, PyTest, Release Automation, Observability, Metric Pipelines
AI & Computer Vision: Deep Learning, Semantic Segmentation, HRNetV2, CARLA

PROFESSIONAL EXPERIENCE

Oracle (OCI)
Software Engineer, OCI Compute Infrastructure | Austin, TX — July 2025 - Present

- Owned Oracle Linux 9 hypervisor image qualification, debugging rollout failures across production infrastructure and distinguishing product defects from test and infrastructure noise to drive release decisions
- Led production deployments across infrastructure spanning 200,000+ hypervisors, maintaining firmware, kernel, and security reliability during critical releases
- Built Release Coordinator Copilot, an AI-assisted release system that synthesizes validation state, failed-test history, release context, and remediation guidance into decision-ready recommendations while preserving human review and accountability
- Extended release validation across 400+ automated test suites and multiple infrastructure repositories, improving consistency and traceability of production reliability decisions
- Designed and implemented automated stress testing for already-running virtual machines, identifying stability regressions introduced by hypervisor image updates that startup testing could not detect
- Delivered FIPS-compatible RPM artifacts for x86_64 and aarch64, validating package integrity and cross-architecture compatibility for Oracle Linux

Software Engineer Intern Oracle Object Storage | Seattle, WA — June - Aug 2024

- Designed and deployed a metrics pipeline ingesting 50+ tenant-level signals from distributed object storage replication and copy services for network and latency analysis
- Authored 50+ MQL-based monitors to isolate replication and networking issues, reducing fault localization time by ~40%

Software Engineer Intern Oracle Virtual Machine Infrastructure | Austin, TX — May - Aug 2023

- Built a real-time server power-supply fault detection system, reducing mean detection time by ~60%, and enabling live repair services
- Verified fault scenarios across 3+ hardware generations, supporting migration to non-terminating virtual machine repair models

Carleton College | Northfield, MN
Computer Science Course Staff and Lab Assistant — Sept 2022 - Sep 2024

- Mentored 50+ students in algorithms, computer systems, and computer architecture
- Supported coursework on pipelining, cache design, and low-level performance reasoning

PROJECTS

Computer Vision For Autonomous Driving | *Undergraduate Thesis* — Sep 2024 - Mar 2025

- Generated 1M+ multimodal simulation samples using CARLA with custom weather, vehicle density, and sensor configurations
- Trained and evaluated HRNetV2 semantic segmentation models under weather-driven domain shift, recovering up to 95% mIoU and reducing fog-induced performance drops by ~35%

RoleFit Platform | *Personal Project* — May 2026

- Built a local Python platform for ATS job-feed ingestion, deterministic role scoring, SQLite-backed tracking, and browser-based workflow automation
- Implemented resume matching and tailoring pipelines with US/remote filtering, ATS deduplication, status workflows, and formatted DOCX export

Scheme Interpreter | *Personal Project* — Dec 2023

- Built a complete interpreter supporting primitives and continuations, strengthening parser, evaluator, runtime, and systems-level design fundamentals

EDUCATION

University of Texas at Austin — Expected Dec 2028
Master of Science in Computer Science

Carleton College — Graduated 2025
B.A. Computer Science, Concentration in Mathematics & Finance | GPA 3.72
Activities: Varsity Football, Club Rugby (**Forwards Captain & Treasurer**), Venture Capital Carleton (**VP**)

CERTIFICATIONS & AWARDS

Certifications: Oracle Cloud Infrastructure Certified Foundations Associate
Awards: 2023 CSC Academic All-District, 2022-2024 NCR Scholastic All-American, 2022-2024 MIAC Academic All-Conference